IMAGE HANDLING AND CANNY EDGE DETECTION

AIM

To read an image using Python and OpenCV, and convert the image into an outline by detecting its edges using the Canny function.

PROCEDURE

1. Install Python and the OpenCV library.

2. Place the input image (sample.jpg) in the same folder as the Python program.

3. Read the image using cv2.imread().

4. Convert the colour image into grayscale using cv2.cvtColor().

5. Apply cv2.Canny() to detect the edges in the grayscale image.

6. Display the original image and the edge-detected outline using cv2.imshow().

7. Press any key to close the image windows.

PROGRAM

```
import cv2

# Read image
img = cv2.imread("sample.jpg")

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Apply Canny edge detection
edges = cv2.Canny(gray, 100, 200)

# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Canny Outline", edges)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

SAMPLE INPUT



SAMPLE OUTPUT



RESULT

Thus, the input image was successfully read using Python and OpenCV, and its outline was obtained using the Canny edge detection function.